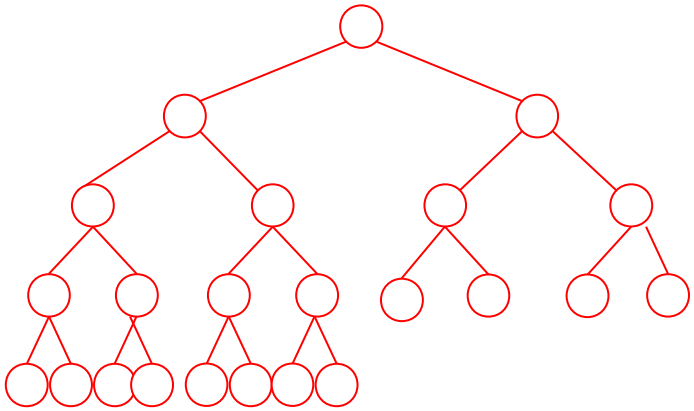
Q1. (30 + 30 = 60pts)

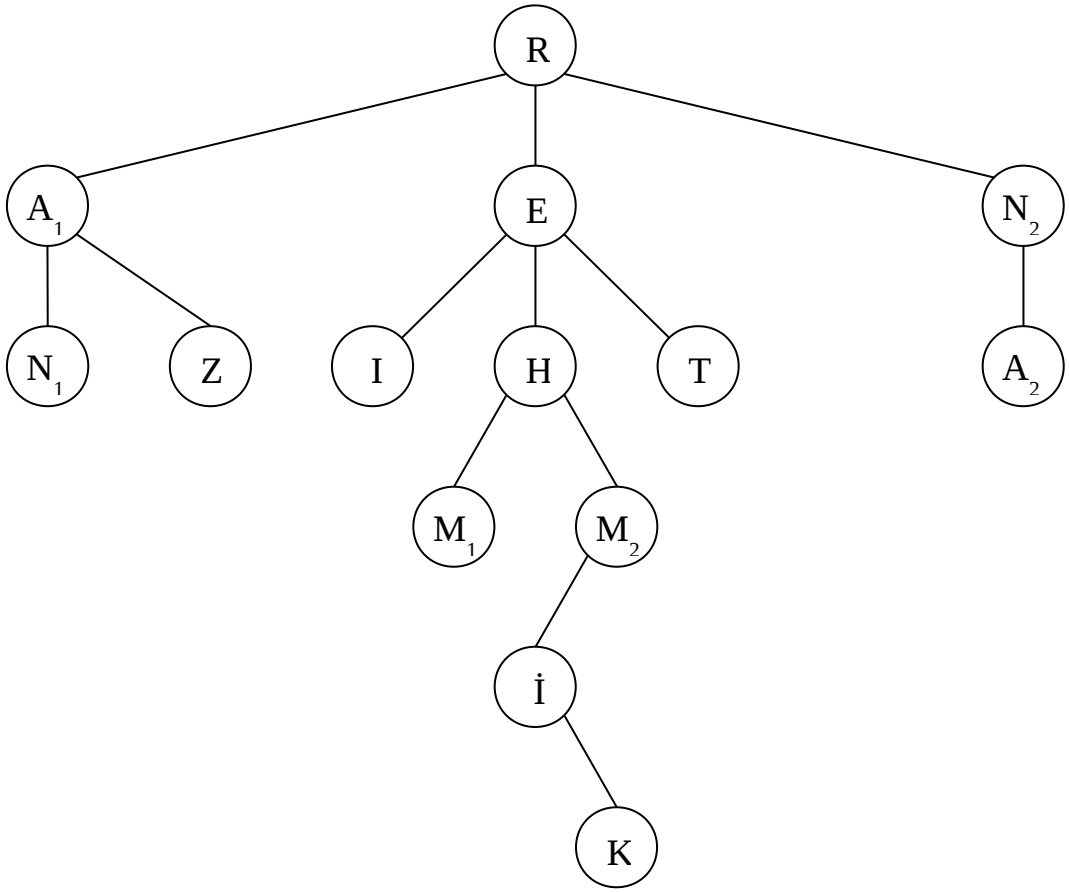
a) Which sequence(s) of nodes below do NOT construct a path at a binary search tree (BST)?

12 64 54 18 26 34 40 35 36 37 38 path
96 84 72 60 36 58 54 38 40 48 44 path
48 54 60 96 88 84 80 76 72 64 56 not path

b) How many different 23-node best BST topologies (i.e., topology with the shortest total access time among all 23-node BST topologies) can you find? Show your figures and calculations that lead to your result!



$$\binom{16}{8} = \frac{16*15*14*13*12*11*10*9}{8*7*6*5*4*3*2*1} = 12720$$

Q2. (40pts)

Place relevant letters in the nodes of the following ternary tree such that an LMNR (i.e., left-middle-node-right) traversal on the tree will result in "$N_1A_1ZIM_1$ HİKM_2ET RA_2N_2" to be displayed.